

MEGAVARUN N

Senior Software Engineer · Payments Infrastructure & FinTech

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PROFESSIONAL SUMMARY

Senior software engineer with 4+ years building production payment infrastructure end-to-end — backend microservices, Android terminal software, and mobile apps. Develop and maintain the platform managing the payment terminal fleet, built full-stack offline card payments, and integrated multiple payment acquirers including L3 network certification.

WORK EXPERIENCE

Surfboard Payments Pvt. Ltd.

Senior Software Engineer

May 2022 – Present

Chennai, Tamil Nadu

Payments platform — Node.js/TypeScript backend microservices, Android (Kotlin) terminal software, and Flutter (Dart) mobile apps.

Platform & Terminal Services

- **Terminal Management Service (TMS):** Develop and maintain the service managing the full lifecycle of the terminal fleet — registration, over-the-air (OTA) software updates, and layered configuration (NestJS, PostgreSQL, Redis, MQTT); ran a zero-downtime migration of the live terminal-events table.
- **Software review & signing pipeline:** Built two services automating the checking, review, and signing of apps deployed to terminals — an audited review-and-approval workflow, and a background worker on GCP Kubernetes running compatibility checks, automated reviews, and app signing.
- **Terminal metrics pipeline:** Built real-time capture of terminal health data (battery, signal, network type) across the terminal fleet, with scheduled nightly aggregation into analytics, avoiding heavy per-event database load.
- **Platform maintenance:** Maintain the core payments acquiring gateway and terminal-services platform (registration, configuration, software distribution, monitoring); review teammates' code for correctness, security, and consistency; fixed a race condition in the shared messaging library that cut duplicate events by ~60–65%.

Android Terminal & Offline Payments

- **Offline card payments (full-stack, L3 certified):** Built offline card payments across terminal, app, and backend for low-connectivity venues — secure on-device transaction storage, offline approval handling to raise approval rates, a crash-safe queue that recovers interrupted payments on restart, and backend settlement; **170k+** offline transactions processed across live-event and festival venues.

Acquirer Integrations

- **Fiserv — online transaction integration (L3 support):** Built the online integration with Fiserv's IPG (Integrated Payments Gateway) — authorization, capture, and refunds with secure card-data encryption; supported L3 certification and debugged scheme test-case failures.
- **Tribe Payments — new acquirer integration (L3 certified):** Built the Tribe integration on Tribe's TAPI protocol in the payments gateway (Node.js/TypeScript) — payload encryption, card-data processing, PIN security, and refunds; integrated into mobile softPOS and Apple Tap-to-Pay; carried out full L3 certification on payment terminal and Android softPOS.

SKILLS

Languages: TypeScript, JavaScript, Kotlin, Dart, SQL, HTML/CSS

Backend: Node.js, Bun, NestJS, Fastify, Express, REST APIs

Mobile: Flutter, Android (Kotlin), Platform Channels

Payments: EMV, PIN Security, Card Encryption, DUKPT, softPOS, Apple Tap-to-Pay, L3 Network Certification

Data & Messaging: PostgreSQL, MySQL, MongoDB, SQLite, Redis, Redis Streams, MQTT

Cloud & DevOps: GCP (GKE), Kubernetes, Docker, AWS, Firebase, GitLab CI/CD, OTA Updates

Practices: Microservices, Event-Driven Architecture, System Design, Distributed Systems, Automated Testing, Code Review

EDUCATION

B.Tech — Information Technology

2016 – 2020